Study of System Design Concepts and Development Tools

Date: 15.07.2026
Ex. No: 1

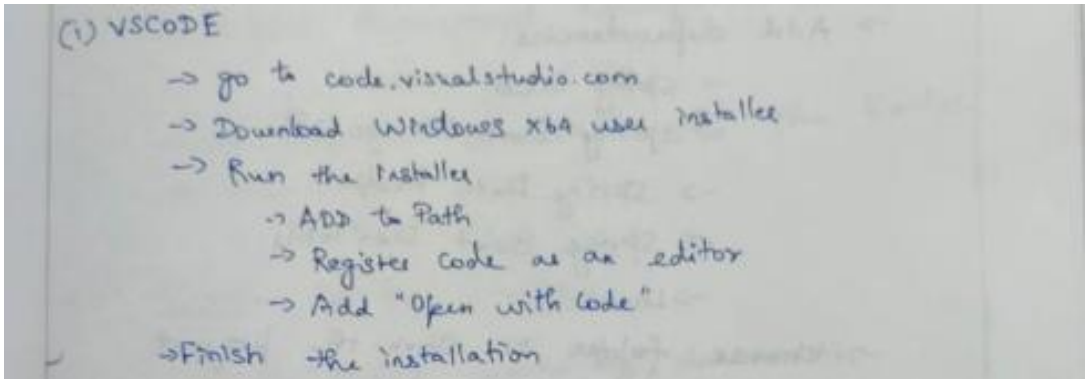
Name: Amrit Gopinath
Reg No: 3122 24 5001 015

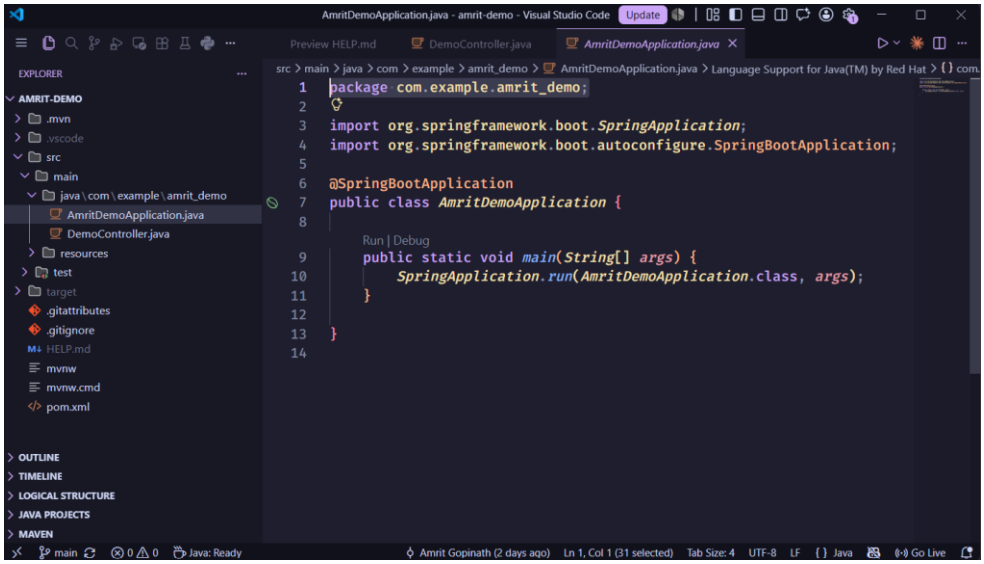
1.Study the fundamentals of System Design and explore the installation, configuration, and applications of the software tools used in the System Design Laboratory.

2.Explore the purpose of each software tool used in the System Design Laboratory and prepare a comparison table indicating its role in developing scalable distributed systems.

3.Create a GitHub repository, configure a Java project in Visual Studio Code, and upload the project to GitHub.

Installation and Setup of different software for system-design:

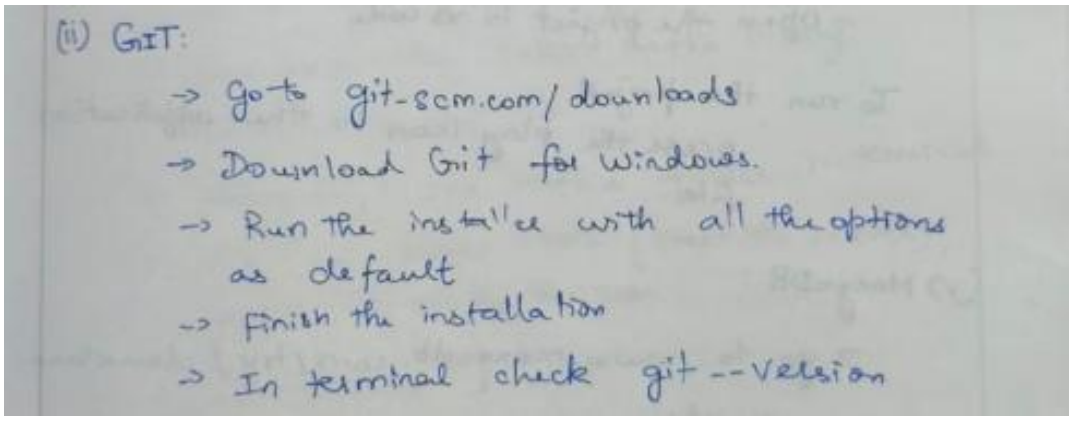




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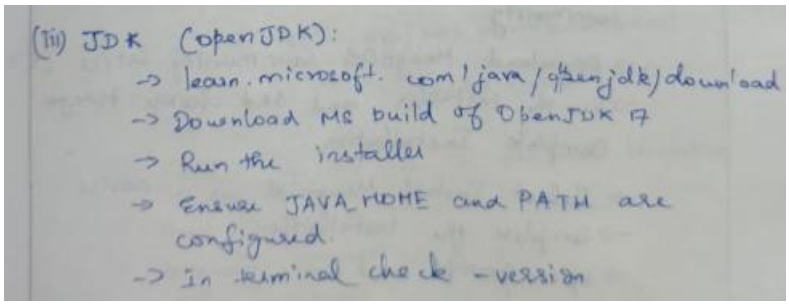
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Installation Proof:

```
Loading personal and system profiles took 735ms.
amrit@AMRITG
~ ❯ git --version
git version 2.50.1.windows.1
amrit@AMRITG
~ ❯
```



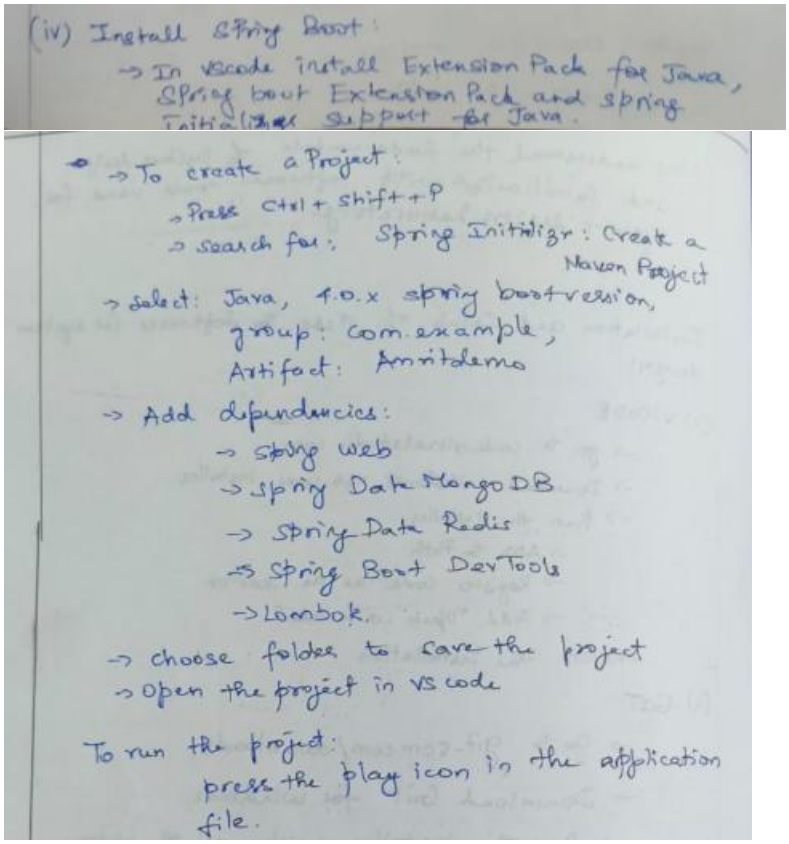
Installation Proof:

```
amrit@AMRITG                                                    ✓ took ⧗ 14
~ ❯ java --version
openjdk 17.0.17 2025-10-21 LTS
OpenJDK Runtime Environment Microsoft-12574423 (build 17.0.17+10-LTS)
OpenJDK 64-Bit Server VM Microsoft-12574423 (build 17.0.17+10-LTS, mixed mode, sharing)
amrit@AMRITG                                                    ✓ took ⧗ 18
~ ❯
```

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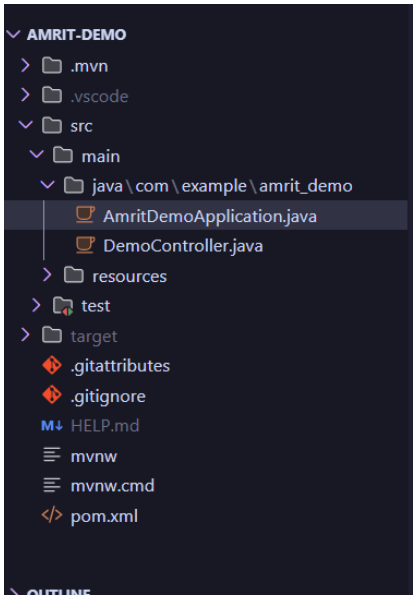


(iv) Install Spring Boot:
-> In vscode install Extension Pack for Java, Spring boot Extension Pack and spring Initializr support for Java.

-> To create a Project:
-> Press Ctrl + shift + P
-> search for: Spring Initilizr: Create a Maven Project
-> Select: Java, 4.0.x spring boot version,
group: com.example,
Artifact: Amritdemo
-> Add dependencies:
-> Spring Web
-> Spring Data MongoDB
-> Spring Data Redis
-> Spring Boot DevTools
-> Lombok.
-> choose folder to save the project
-> Open the project in VS code

To run the project:
press the play icon in the application file.

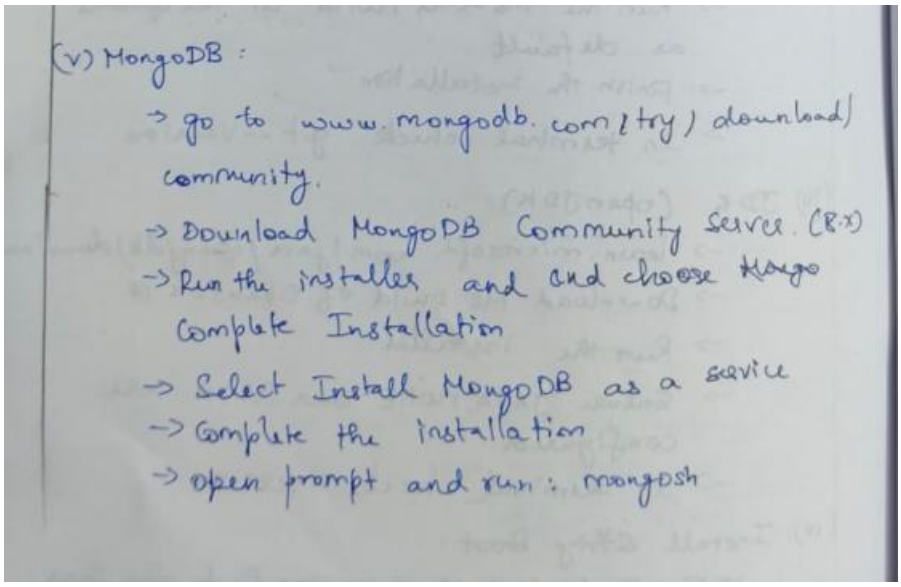
Project Structure:



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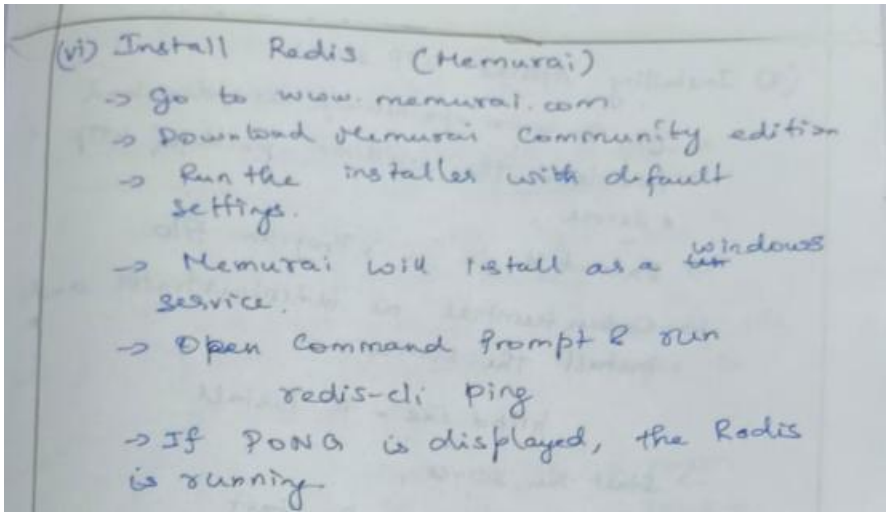
Installation Proof:

```
amrit@AMRITG                                                    × took ⧗ 84ms  21:
~ ❯ mongosh
Current Mongosh Log ID: 6a5a57264648778e0b628c9f
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=m
.6.0
Using MongoDB:          8.2.3
Using Mongosh:          2.6.0
mongosh 2.7.0 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

   The server generated these startup warnings when booting
   2026-07-12T09:42:03.592+05:30: Access control is not enabled for the database. Read and write access to data
iguration is unrestricted

test>
```



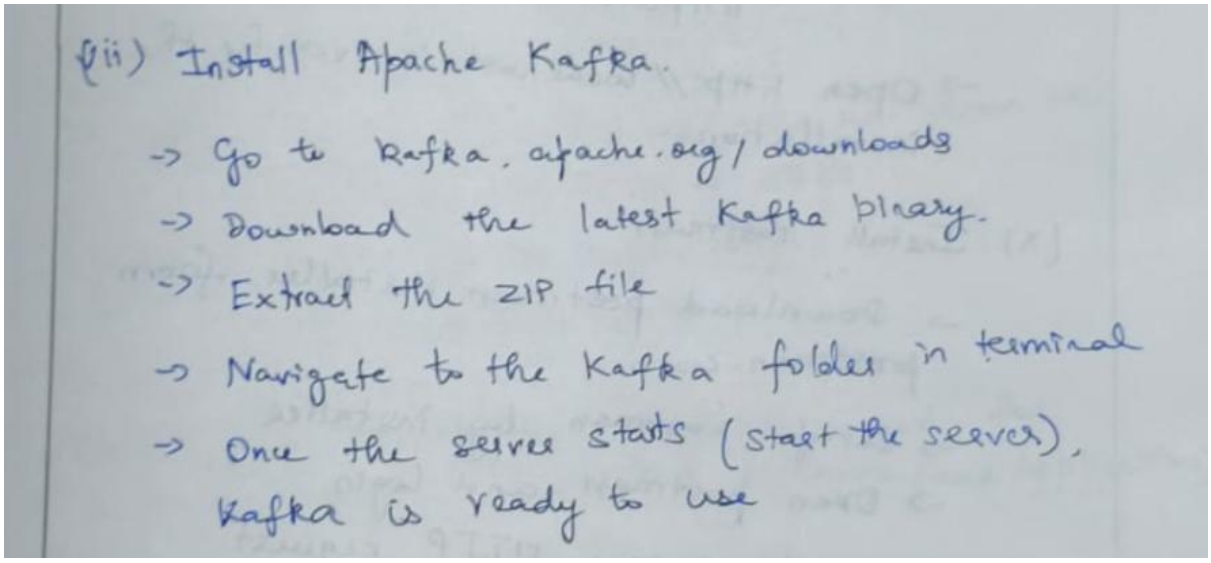
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Installation Proof:

```
amrit@AMRITG
~ ❯ memurai-cli ping
PONG
```



Installation Proof:

```
amrit@AMRITG:~$ cd kafka/
amrit@AMRITG:~/kafka$ bin/kafka-console-consumer.sh --topic test-topic --bootstrap-server localhost:9092 --from-beginning
The consumer rebalance protocol (KIP-848) is production-ready! Set group.protocol=consumer to try it out. See https://kafka.apache.org/documentation/#consumer_rebalance_protocol
bin/kafka-console-consumer.sh --topic test-topic --bootstrap-server localhost:9092 --from-beginning



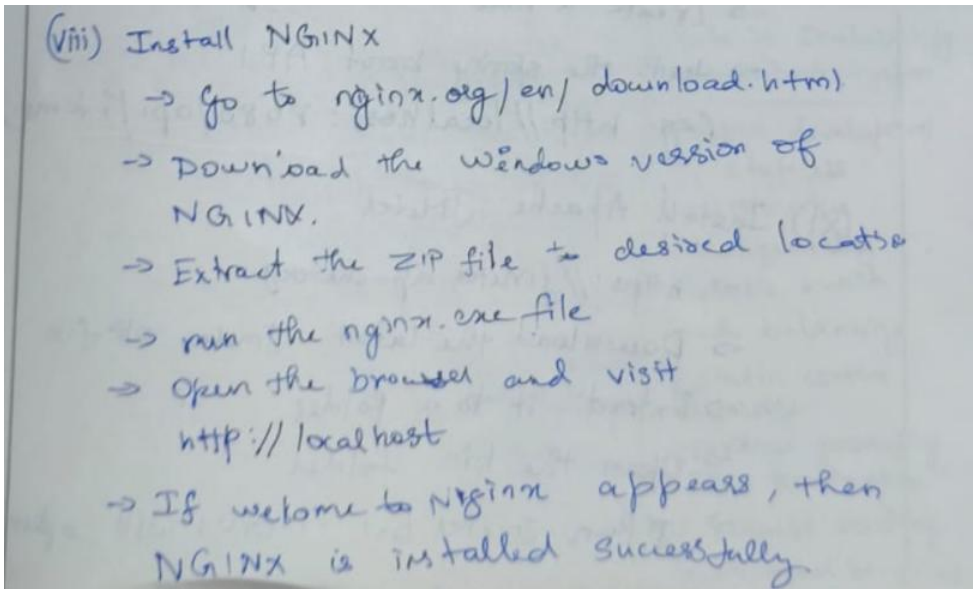
;
test
Hell world
testing kafka
```

```
amrit@AMRITG:~$ tar -xzf kafka_2.13-4.3.1.tgz
amrit@AMRITG:~$ mv kafka_2.13-4.3.1 kafka
amrit@AMRITG:~$ cd kafka
amrit@AMRITG:~/kafka$ KAFKA_CLUSTER_ID="$(bin/kafka-storage.sh random-uuid)"
amrit@AMRITG:~/kafka$ bin/kafka-storage.sh format -t $KAFKA_CLUSTER_ID -c config/server.properties
Because controller.quorum.voters is not set on this controller, you must specify one of the following: --standalone, --initial-controllers, or --no-initial-controllers.
amrit@AMRITG:~/kafka$ bin/kafka-storage.sh format -t $KAFKA_CLUSTER_ID -c config/server.properties --standalone
Bootstrap metadata: BootstrapMetadata(records=[ApiMessageAndVersion(FeatureLevelRecord(name='metadata.version', featureLevel=30) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='eligible.leader.replicas.version', featureLevel=1) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='group.version', featureLevel=1) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='share.version', featureLevel=1) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='streams.version', featureLevel=1) at version 0), ApiMessageAndVersion(FeatureLevelRecord(name='transaction.version', featureLevel=2) at version 0)], metadataVersionLevel=30, source=format command)
Formatting dynamic metadata voter directory /tmp/kraft-combined-logs with metadata.version 4.3-IV0.
amrit@AMRITG:~/kafka$ bin/kafka-server-start.sh config/server.properties
[2026-07-17 14:22:55,317] INFO Registered `kafka:type=kafka.Log4jController` MBean (kafka.utils.Log4jControllerRegistration$)
[2026-07-17 14:22:55,755] INFO Registered signal handlers for TERM, INT, HUP (org.apache.kafka.common.utils.LoggingSignalHandler)
[2026-07-17 14:22:55,759] INFO [ControllerServer id=1] Starting controller (kafka.server.ControllerServer)
[2026-07-17 14:22:56,427] INFO Updated connection-accept-rate max connection creation rate to 2147483647 (kafka.network.ConnectionQuotas)
[2026-07-17 14:22:56,479] INFO [SocketServer listenerType=CONTROLLER, nodeId=1] Created data-plane acceptor and processors for endpoint : ListenerName(CONTROLLER) (kafka.network.SocketServer)
[2026-07-17 14:22:56,487] INFO authorizerStart completed for endpoint CONTROLLER. Endpoint is now READY. (org.apache.kafka.server.network.Endpoi
```

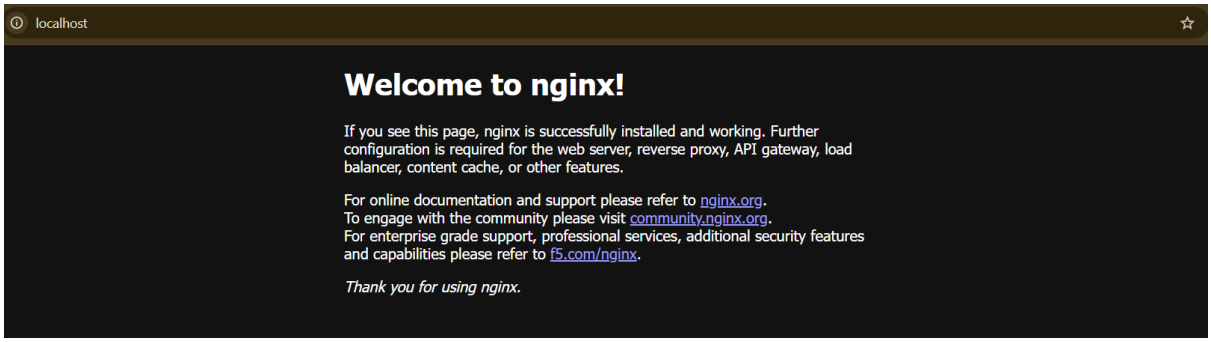
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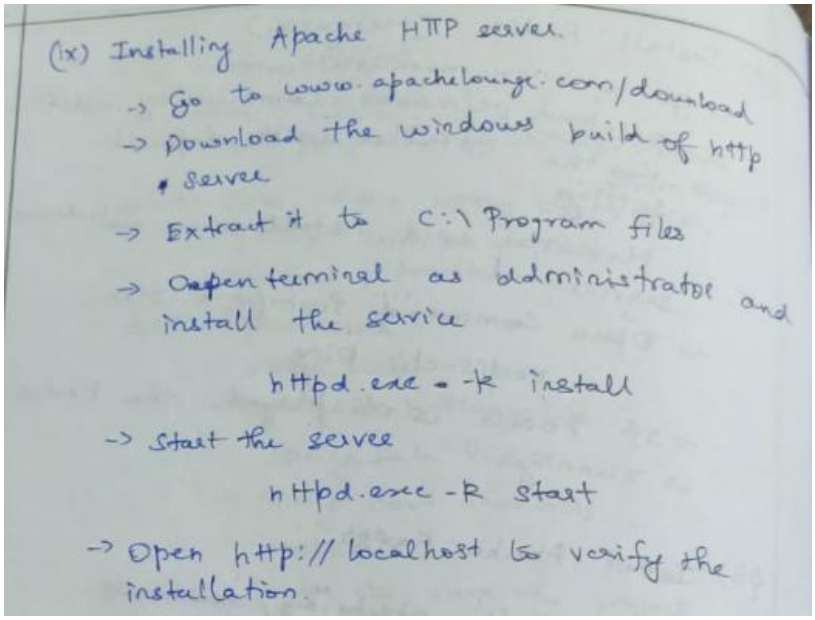
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Installation step:





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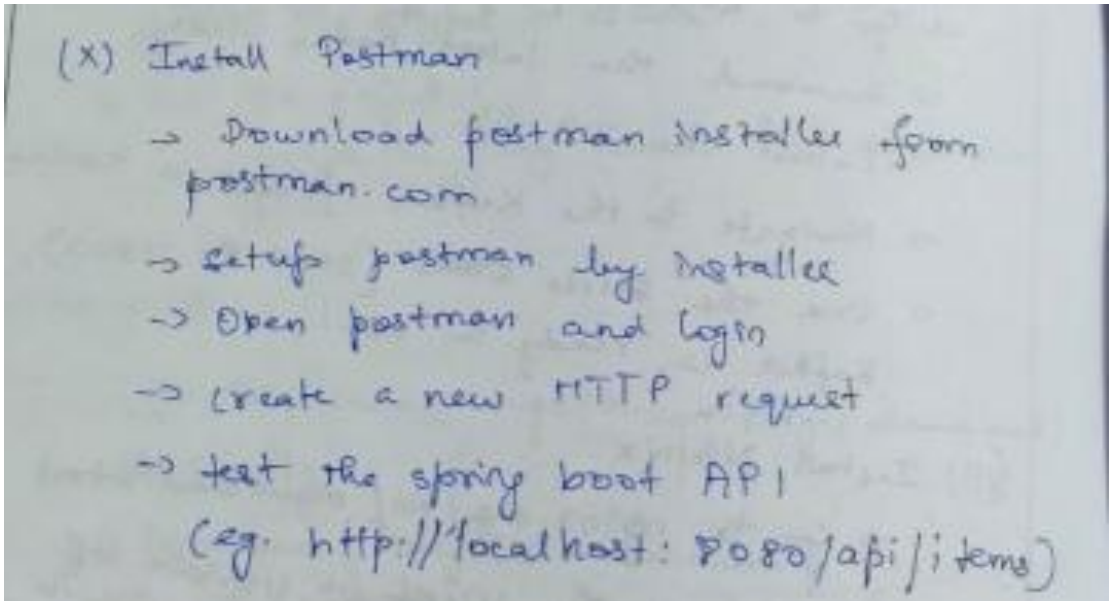
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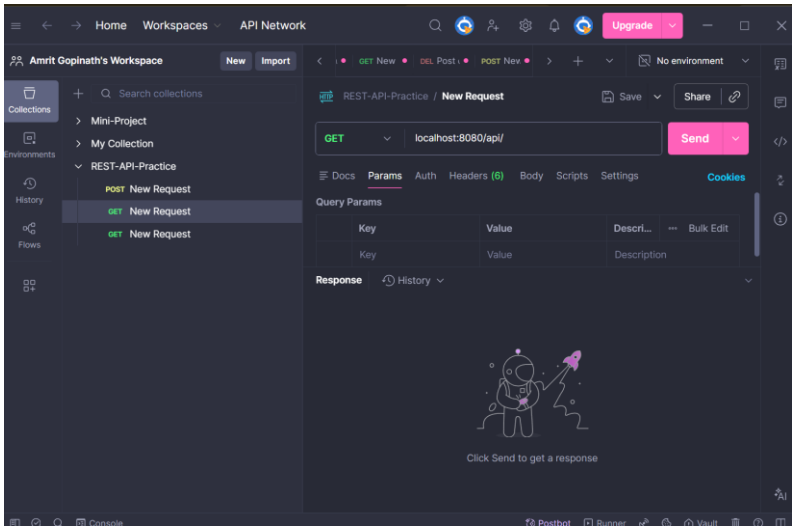
Installation step:



It works!



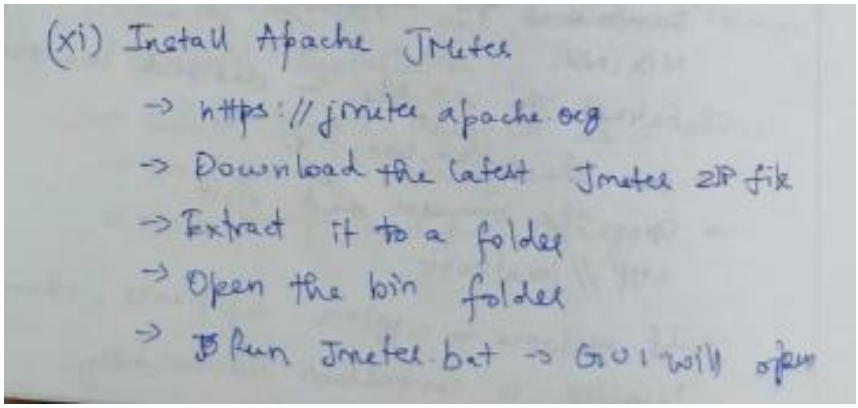
Installation step:



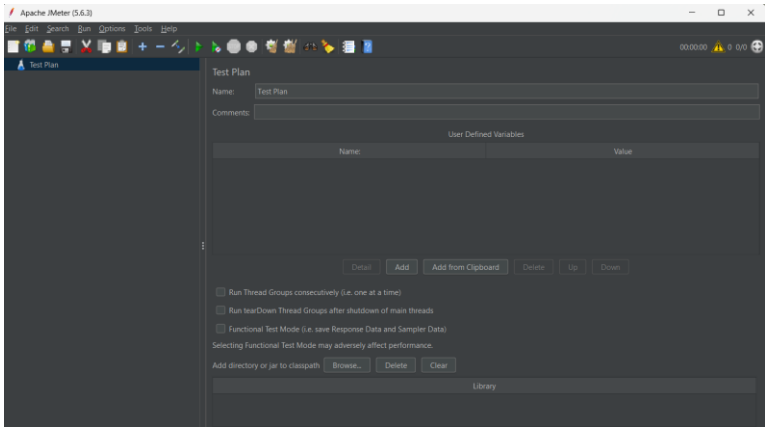
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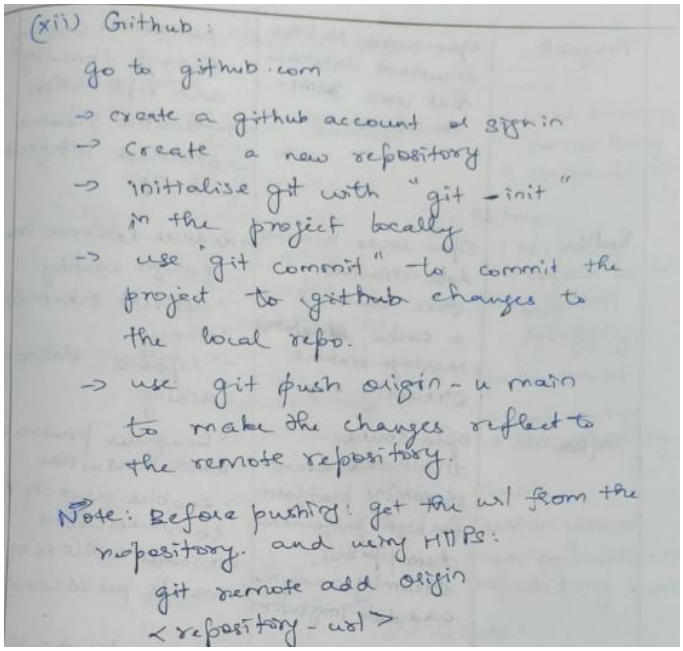
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Installation step:

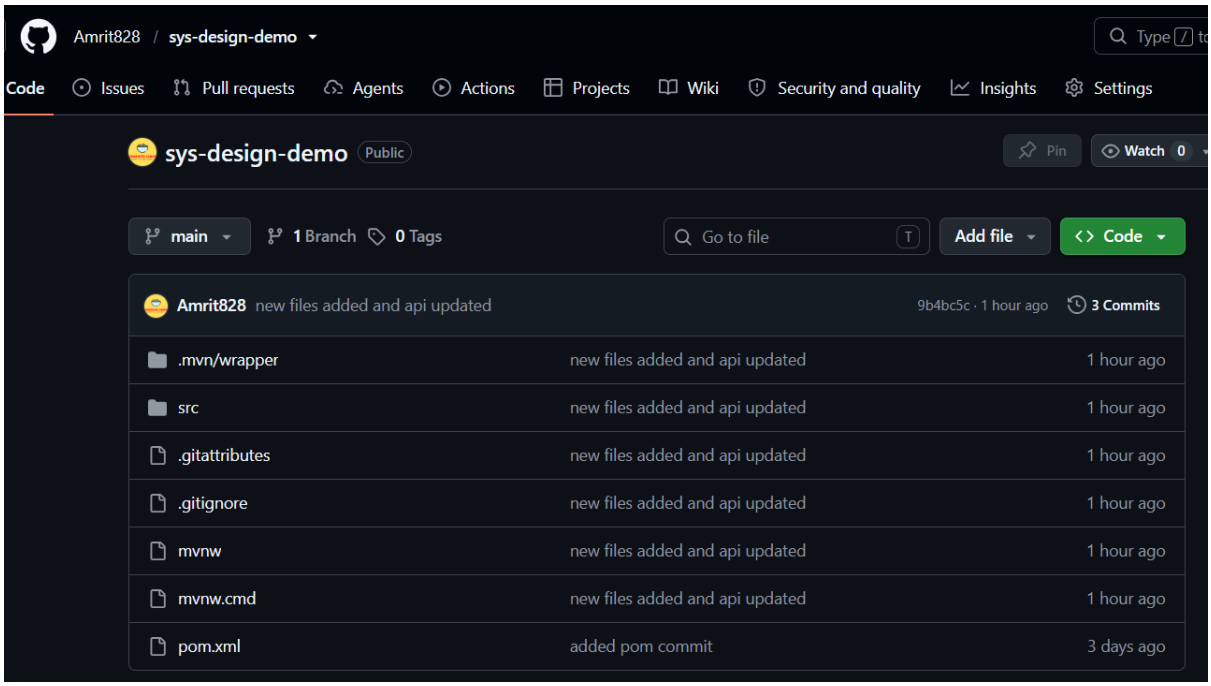




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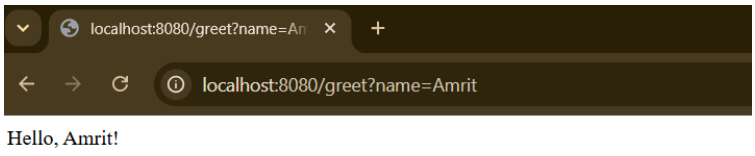
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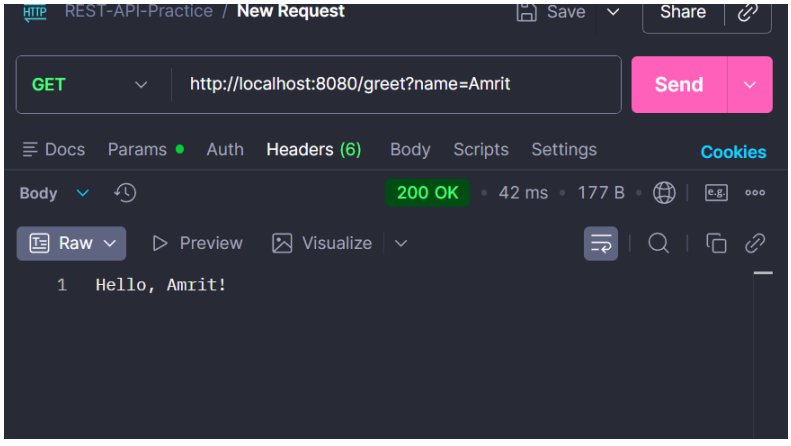
3) Github Link: https://github.com/Amrit828/sys-design-demo

Demo Output:



Hello, Amrit!

Postman Output:

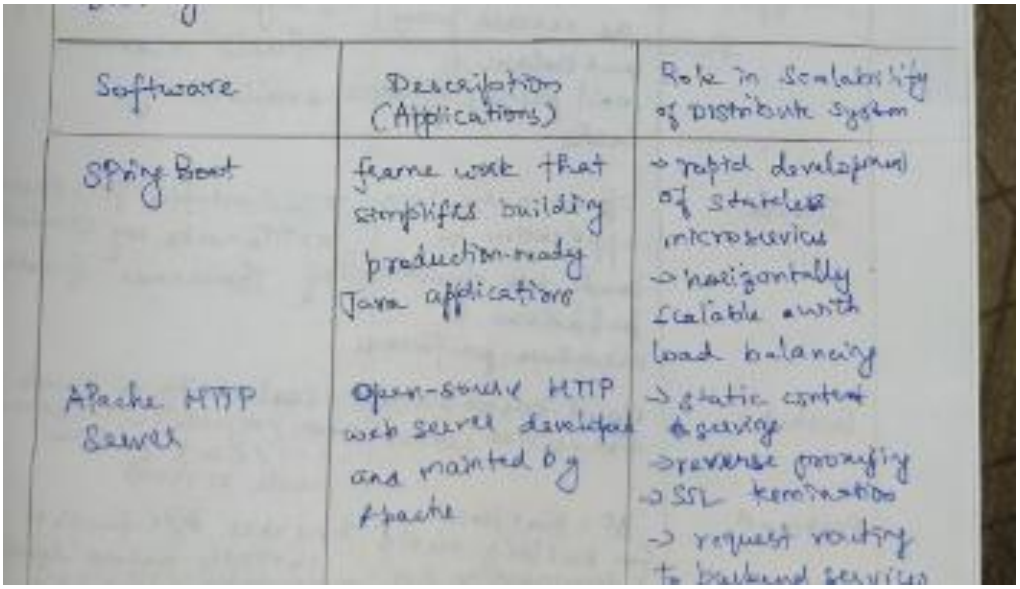


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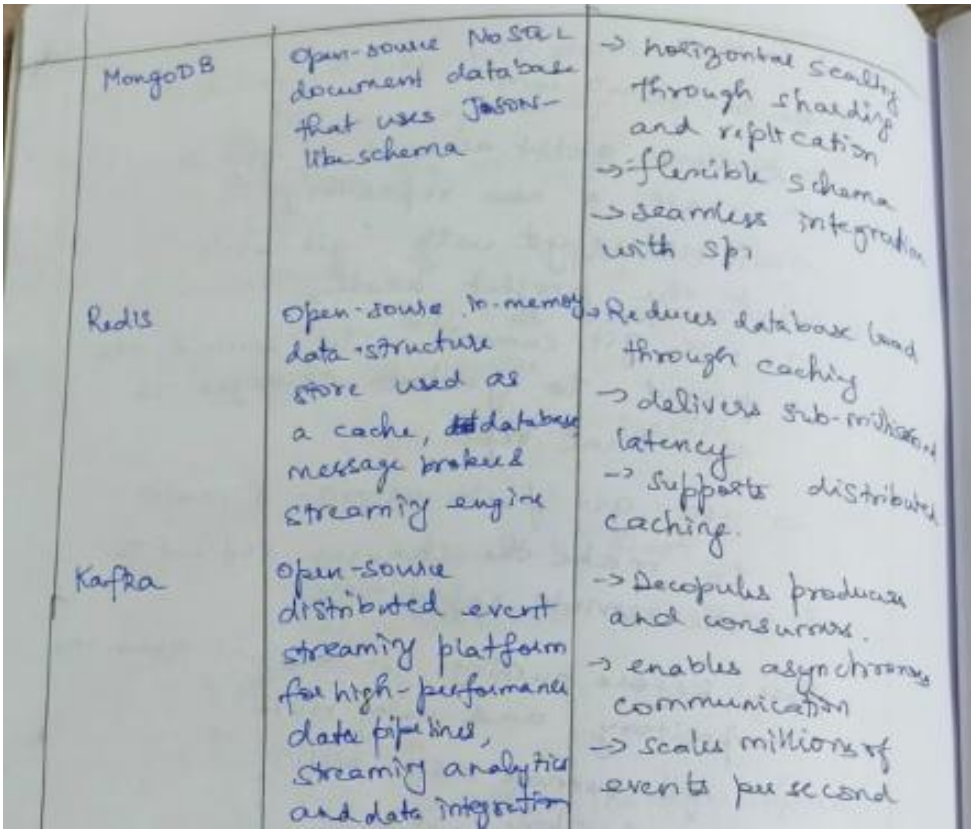
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Comparison of different software for developing scalable and distributed systems:



Software	Description (Applications)	Role in Scalability of distribute system
Spring Boot	framework that simplifies building production-ready Java applications	→ rapid development of stateless microservices → horizontally scalable with load balancing
Apache HTTP Server	Open-source HTTP web server developed and maintained by Apache	→ static content serving → reverse proxying → SSL termination → request routing to backend services

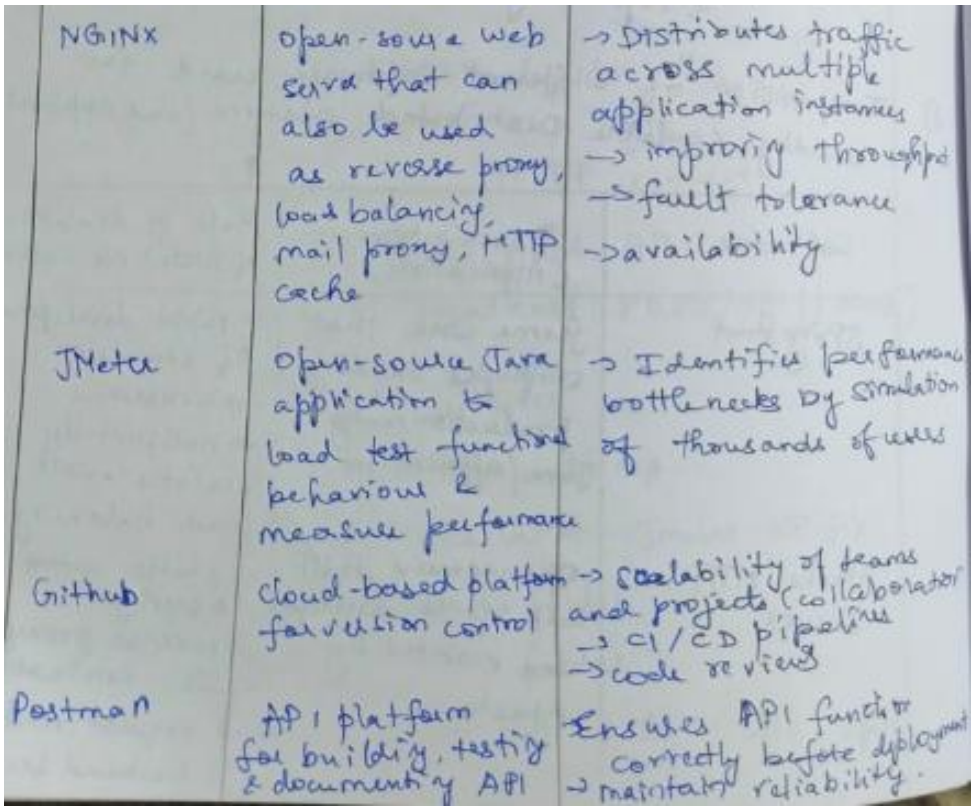


MongoDB	Open-source NoSQL document database that uses JSON-like schema	→ horizontal scaling through sharding and replication → flexible schema → seamless integration with Spr
Redis	Open-source in-memory data-structure store used as a cache, database, message broker & streaming engine	→ Reduces database load through caching → delivers sub-millisecond latency → supports distributed caching.
Kafka	Open-source distributed event streaming platform for high-performance data pipelines, streaming analytics and data integration	→ Decouples producers and consumers. → enables asynchronous communication → scales millions of events per second

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NGINX	open-source web server that can also be used as reverse proxy, load balancing, mail proxy, HTTP cache	-> Distributes traffic across multiple application instances -> improving throughput -> fault tolerance -> availability
JMeter	Open-source Java application to load test functional behaviour & measure performance	-> Identifies performance bottlenecks by simulation of thousands of users
Github	Cloud-based platform for version control	-> Scalability of teams and projects (collaborator) -> CI/CD pipelines -> code reviews
Postman	API platform for building, testing & documenting API	Ensures API functions correctly before deployment -> maintains reliability.

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Learning Outcomes:

- Explain the fundamentals of system design.
- Describe the role of major components in scalable distributed systems.
- Identify the purpose of each software tool used in the laboratory.
- Set up the development environment required for implementing system design experiments.